

When The *Flower* Stops *Breathing*

Carla Budar

Parsons School of Design

Spring 2026

Design and Technology



Instagram: @carbudar
Linkedin: Carla Budar
Portfolio: www.carlabudar.com



Thesis Statement

How do humans
coexist with the living
environment in a way
that creates *balance*
rather than *destruction*?

We damage habitats and prioritize growth at the expense of long-term stability. Yet, our survival is tied directly to a healthy environment: when the ecosystem suffers, human life suffers with it.

When The Flower Stops Breathing is an interactive kinetic sculpture composed of a cluster of mechanical flowers that continuously bloom and close, echoing the subtle rhythms of living ecosystems. It asks a simple but uneasy question: what happens to a system when it is constantly observed, approached, and disturbed?

The kinetic form translates environmental fragility into real-time response rather than static representation. Rather than treating ecological decline as distant or abstract, it makes it immediate, something that reacts to presence, attention, and time.

The Interaction

The sculpture responds to human proximity and duration of engagement. At a balanced distance, the flowers move in a calm, rhythmic cycle—open, close, breathe.

But when presence becomes prolonged or too intense, the system begins to shift. Movement becomes irregular. Timing slips. The “breathing” loses stability.

Interaction is not a trigger, it is a condition. The environment changes based on how it is occupied.

How can humans *recognize*
their *presence* within nature as
an *active force* rather than a
distant *observer*?



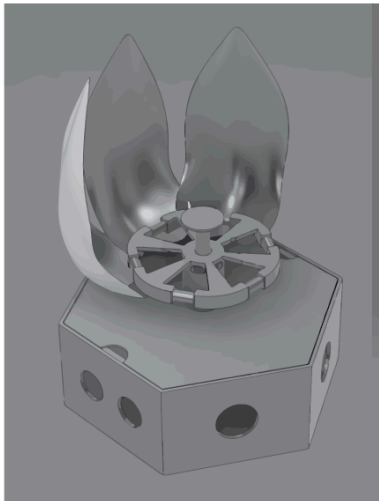
The Process

The process began with sketching ideas by hand (refer to “A Glimpse Into My Thinking Process”) before moving into measuring and modeling each part in Autodesk Fusion 360. As the form developed, I constantly shifted between digital models, paper prototypes, and 3D printed iterations to test movement, structure, and measurements. The rack and pinion mechanism required extremely precise measurements and spacing, making the process tedious at times, with constant adjustments and returns to the drawing board. After moving through twelve different prototypes, the form finally reached a point that felt reliable and refined.

Throughout the process, the most rewarding part was solving one problem at a time, from motor distance and housing shape to the core mechanism and structural supports, slowly turning the system into something functional and foolproof.



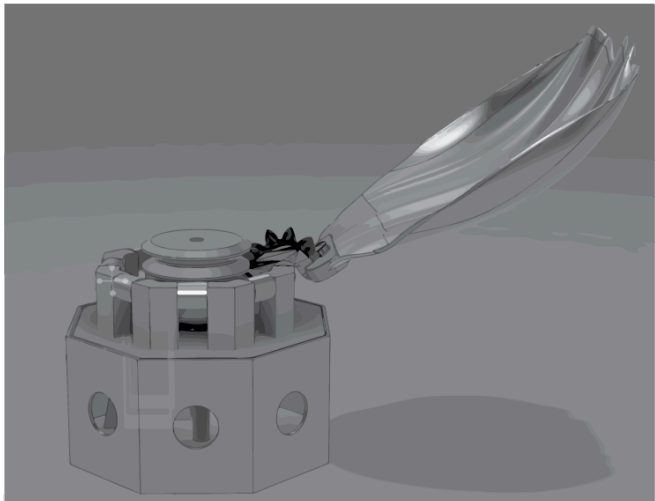
Mechanism *and* Renders



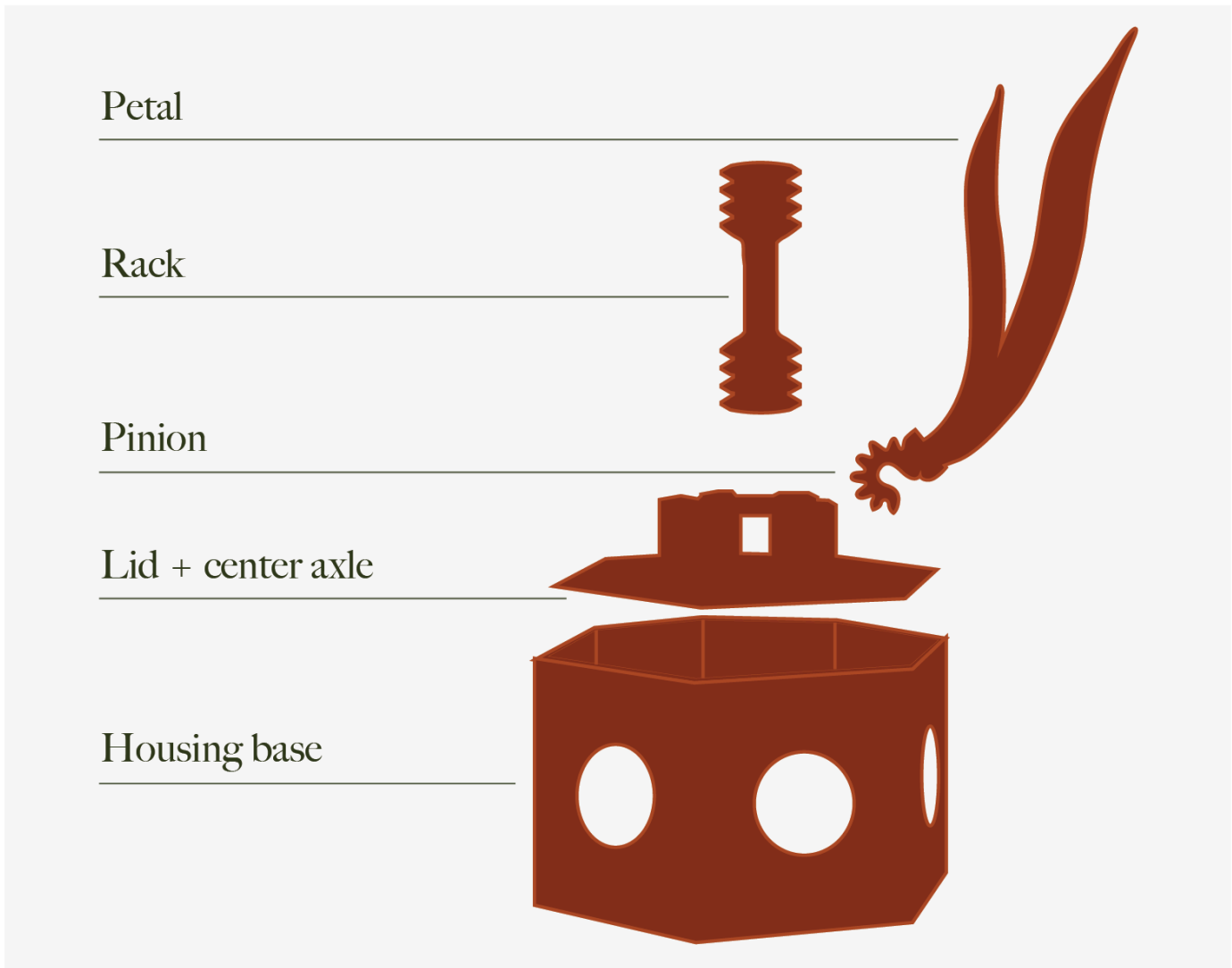
Early Prototype



Final Model (P12)



Big Flower Model





This project is not meant to prevent interaction, but to *reveal its effect*.

The viewer becomes a *variable* within the system, whether intentionally or not. Their presence is *not neutral*; it is a force that *shapes behavior*.

The Goal

The work is aimed at **public audiences** in shared spaces, galleries, installations, or environments where people naturally **gather and observe**. It is especially directed toward viewers who **move quickly** through spaces, encouraging them to slow down and **notice** how **presence alters systems** that appear stable. As viewers attempt to understand what affects the sculpture's behavior, they are invited into **conversation** and collective observation.

At its core, the project communicates **balance: closeness is not harmful, but unconsidered accumulation of presence has consequences**.

The Tech Stack

1. Power Supply

To provide enough power for all the servo motors.

Connected to the servo driver (5).

2. Raspberry Pi

The brain. Receives camera input and sending signals to Arduino Uno (3).

3. Arduino Uno

Controls the servo motor (6) movements and sends signal for when the LEDs (7) are on/off.

4. Camera

Main source of input. Using Python OpenCV to detect human presence.

5. 16 Channel Servo Driver

Groups 16 servo together so it does not take too much pin on the microcontroller.

6. Servo Motors

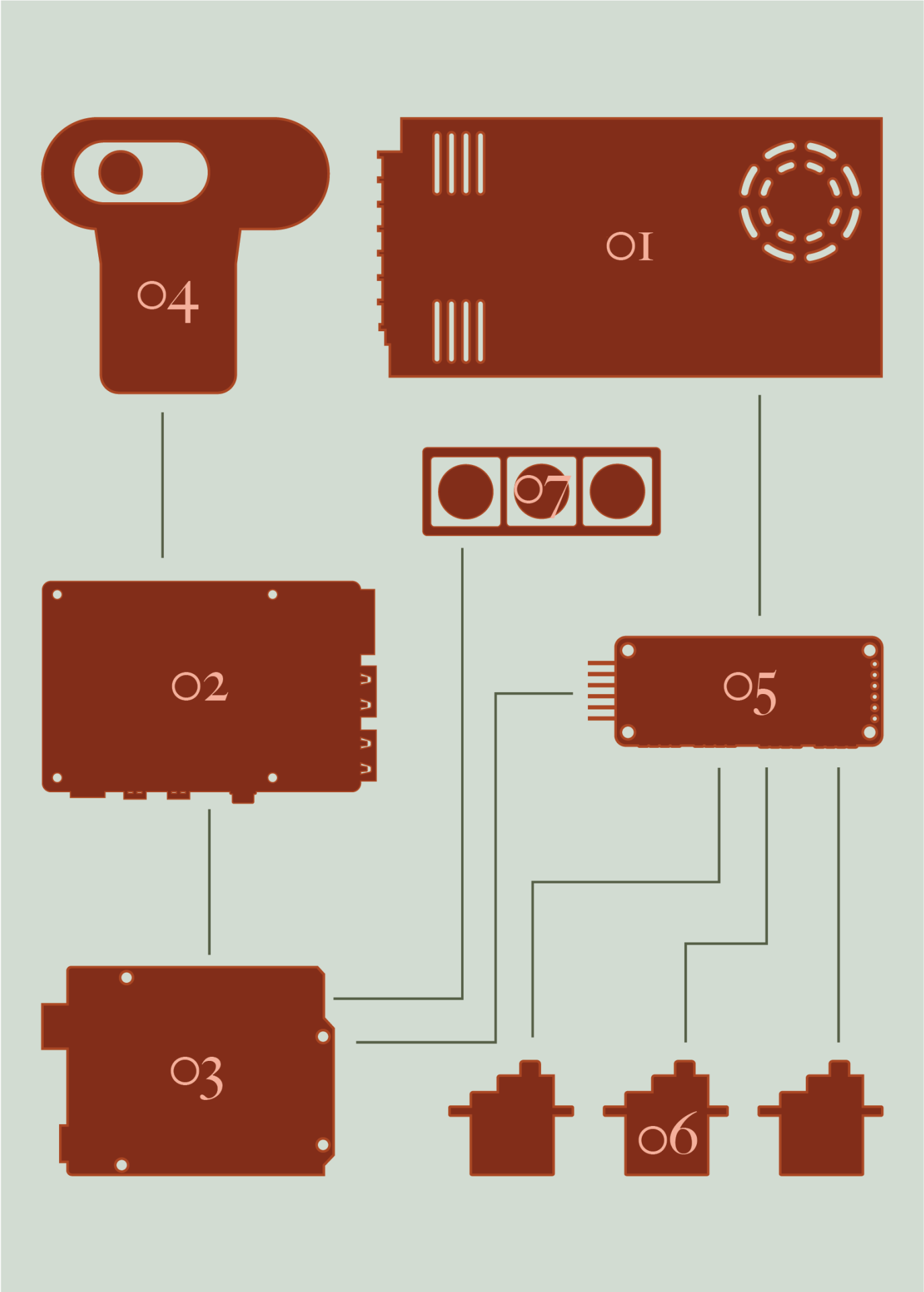
Motor shaft is connected to a spur gear, which converts rotational motion to linear, moving the rack up and down.

7. LED Light

(Addressable LED) Each LED is assigned to a flower and a number. LED behaviour is controlled by the Arduino Uno (3), lighting up when flowers are blooming.

8. 3D Printed Parts

Modeled in Fusion360, all parts, including: petals, housing, rack, and pinion, are 3D printed using PLA filament.



Artist Reflection

Through building this project, I became more aware of how easily systems lose stability when scaled into interaction. What began as a simple mechanical flower quickly turned into a study of sensitivity, how small inputs accumulate into larger behavioral shifts. The most challenging part was not the construction of motion, but the calibration of response: finding the point where interaction becomes disruption.

It was also challenging to communicate the project's intention clearly. A recurring point of feedback centered around the use of manmade materials while discussing themes of environmental fragility and human impact on nature. However, that contrast became an important part of the work itself. The combination of organic floral forms with mechanical movement and plastic materials acts as commentary on the irony of the topic, how something visually delicate and beautiful can still be constructed from artificial materials tied to human advancement and consumption. I wanted the piece to create a sense of discomfort and reflection around that contradiction, while also being mindful that the work itself is preserved rather than treated as something disposable.

This project sits between observation and participation. It made me rethink what it means for something to feel "alive," and how perception changes when behavior is no longer fixed.